DISCOVER · wide-field surveys find O(1000) lensed AGN

CHARACTERIZE

Rubin LSST

pixel 0.2″ · PSF 0.71–0.77″

i r g · 10-yr

Euclid

pixel 0.1–0.3″ · PSF 0.16–0.5″

VIS Y J H · wide

Roman

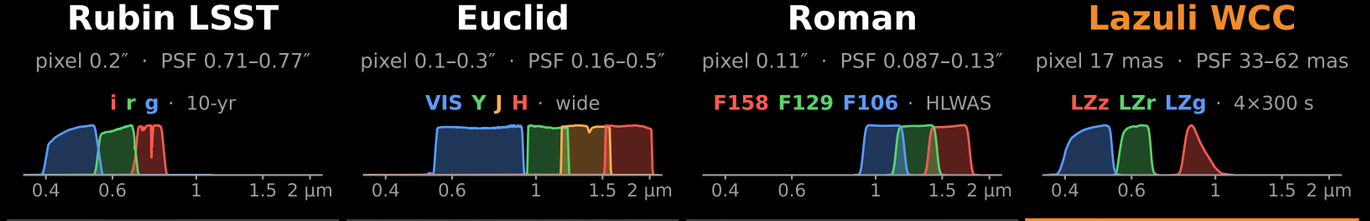
pixel 0.11″ · PSF 0.087–0.13″

F158 F129 F106 · HLWAS

Lazuli WCC

pixel 17 mas · PSF 33–62 mas

LZz LZr LZg · 4×300 s



Quad z_d=0.43 z_s=1.87

Quad z_d=0.89 z_s=2.74

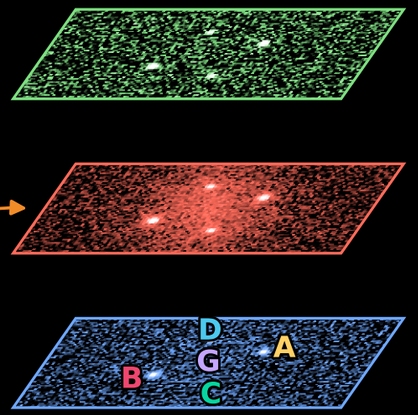
Double z_d=0.57 z_s=1.22

1″

IFS field

LAZULI IFS · one pointing, every image

0.4–1.7 μm · R ≈ 100–500 · 40 mas spaxels · 4 × 900 s



1.40 μm: Hβ line (rest 0.486 μm)

continuum removed: only quasar broad-line gas emits

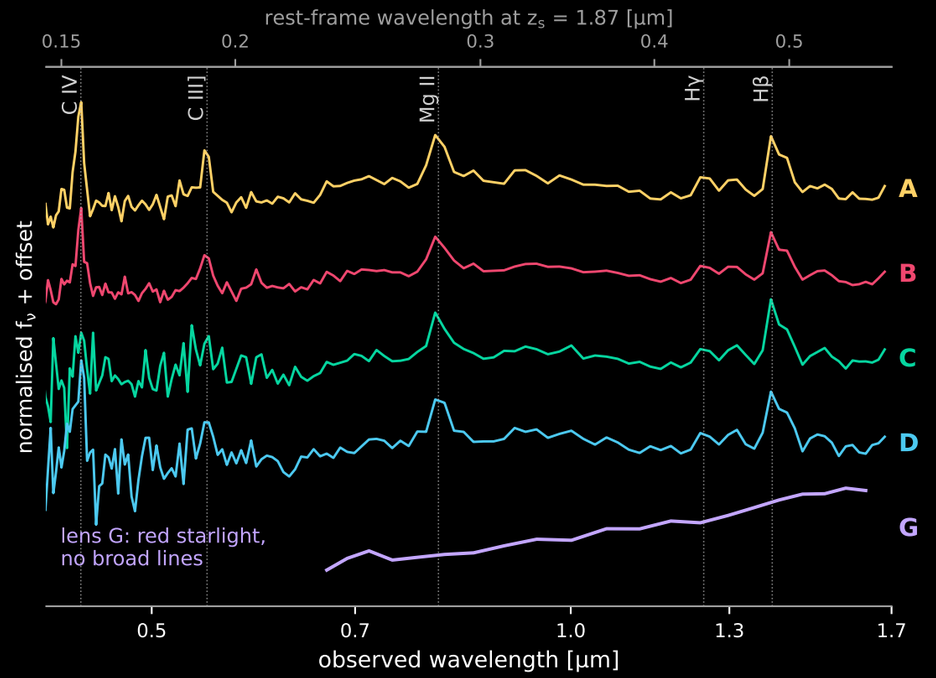
1.00–1.20 μm continuum

red starlight of lens G rivals the quasar images

0.58–0.76 μm continuum

lens starlight fades to the blue; quasars dominate

Same broad lines at the same redshift in every image → lensed quasar confirmed



Simulated with SLSim: COSMOS hosts, qsogen quasar SEDs, static microlensing. Lazuli: 3.06 m, Strehl > 0.8 at 633 nm; specs from Roy et al. 2026 (arXiv:2601.02556). Lazuli throughput, sky, noise and exposure times are notional. Lensed-AGN counts: Oguri & Marshall 2010.